What Drives Flood Insurance Payouts in North Carolina?

CSC 405 individual project, Fall 2026. Robert Grooms.

Research question: What factors predict how much the National Flood Insurance Program pays on a building claim in North Carolina, and do claims in high-risk flood zones actually get paid more than claims in low-risk zones?

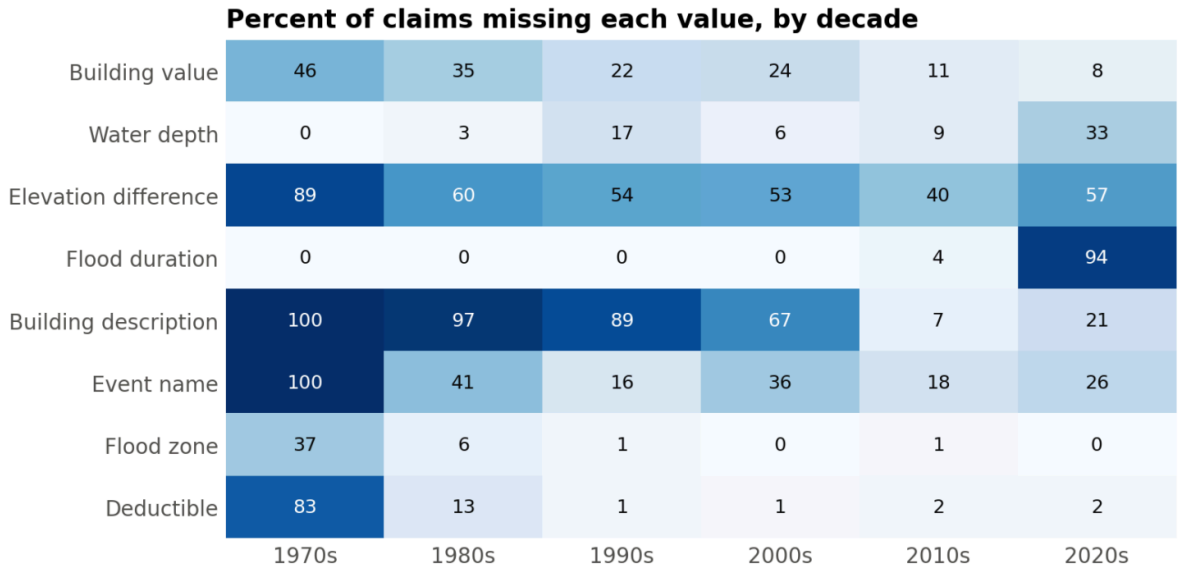
1. Dataset overview

The data is FEMA's NFIP Redacted Claims v3, filtered to North Carolina: **109,593 claims and 84 columns** from 1978 to 2026. Each row is one claim. The target is the net building payment, adjusted to 2025 dollars with the annual Consumer Price Index.

Claims	Paid claims	Median payout
109,589	80,843	$13,822

2. Data cleaning

Missing values



Gaps are tied to time, not scattered at random. Older claims lack the deductible, flood zone and building description. Flood duration is almost empty in the 2020s, so a simple rule such as dropping every

incomplete row would remove whole eras from the analysis.

Missing values also relate to the outcome:

column	pct_missing	paid_rate_missing	paid_rate_present	median_paid_missing	median_
buildingPropertyValue	18.2	0.02	0.9	9297.975	
waterDepth	13	0.68	0.75	8220.98	
elevationDifference	49.6	0.75	0.72	10072.75	
floodWaterDuration	7.6	0.79	0.73	15952.615	
buildingDescriptionCode	52.4	0.7	0.78	6274.305	
floodEvent	21.9	0.67	0.76	5823.65	
ratedFloodZone	1.5	0.62	0.74	7216.62	
buildingDeductibleCode	2.6	0.46	0.75	5106.83	

The clearest case is building value: when it is missing, only 2% of claims were paid, against 90% when it is present. It appears to be recorded mainly when a claim is adjusted, so it is not missing at random. Since the payout model uses paid claims only, this column stays.

Cleaning decisions

Item	Decision	Reason
Negative payments (4)	Removed	They are recoveries, not payouts.
Unpaid claims (26%)	Kept for EDA and tests	Whether a claim is paid is part of the question.
Flood duration	Dropped	94% missing in the 2020s.
Foundation type	Dropped	76% hold a 0, which the data dictionary does not define.
Pre-FIRM flag	Dropped	Contradicts the post-FIRM flag and drifts over time.
Event name	Blank set to Unnamed event	A blank means no named storm.
Flood zone	Grouped into A, V and low risk	Too many detailed codes; V is different from A.
Water depth	Values of 900+ removed	They are placeholders. Units also mix feet and inches.
Coordinates outside NC	Removed	They are data entry errors.
Zero coverage	Treated as missing	Not a real policy amount.
Dollar amounts	Converted to 2025 dollars	Claims span almost 50 years.

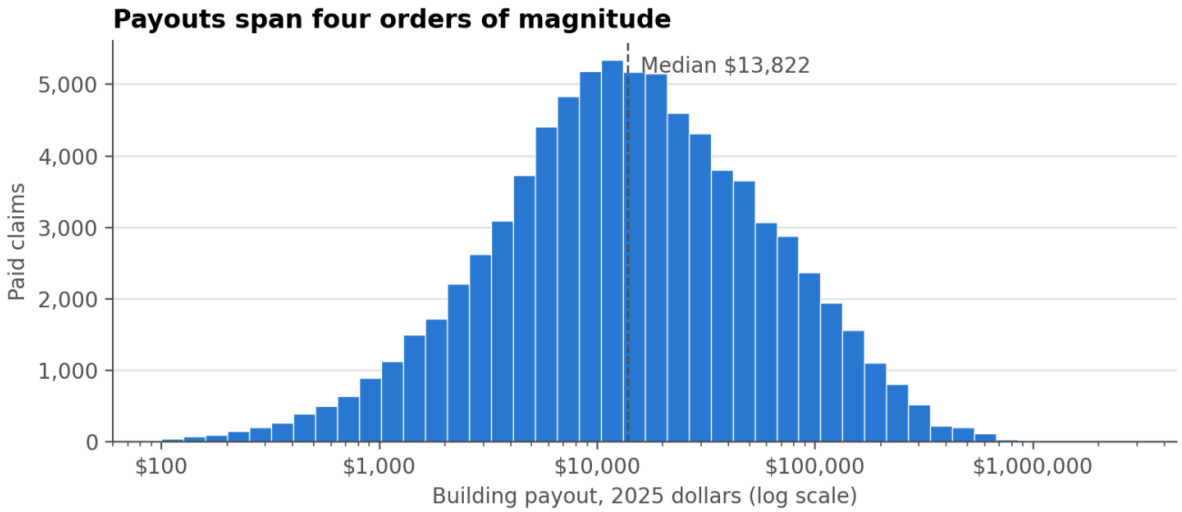
Inflation is adjusted with CPI-U, which tracks consumer prices in general. Building repair costs may have risen at a different rate, so older payouts in 2025 dollars are an estimate.

Outliers

On a log scale, the 1.5 IQR rule flags 50 paid claims above $703,514 and 637 below $291. These are kept: large payouts are real losses and are what insurance is for. Using a log scale limits their effect. Separately, 336 payments exceed the listed coverage. The data does not explain why, so they stay in and are treated as a data quality limit.

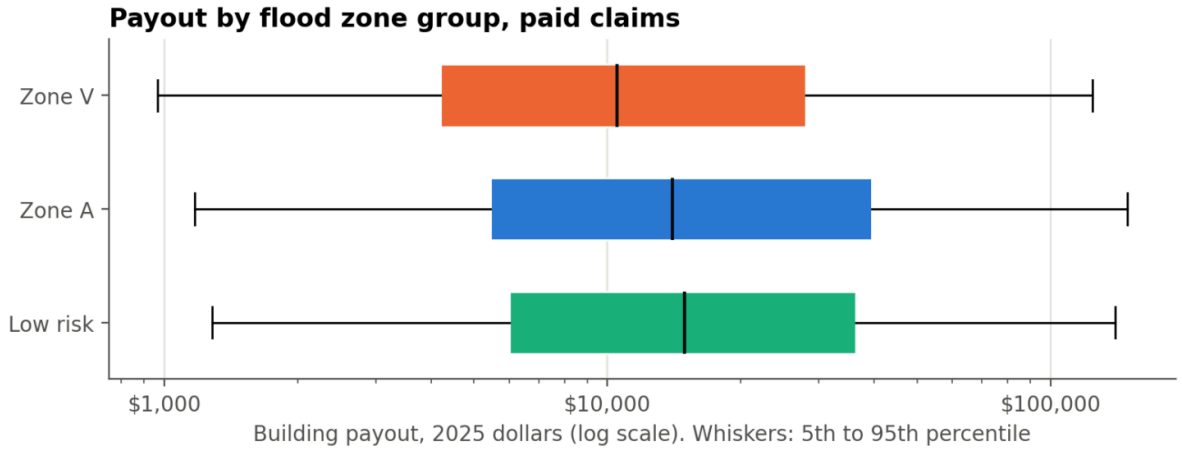
3. Exploratory analysis

Payout distribution



Most paid claims are modest, with a median of $13,822, but the mean is $35,924 and the largest is $2,735,296. The shape is close to a bell curve only on a log scale, so models and tests use the log of payout.

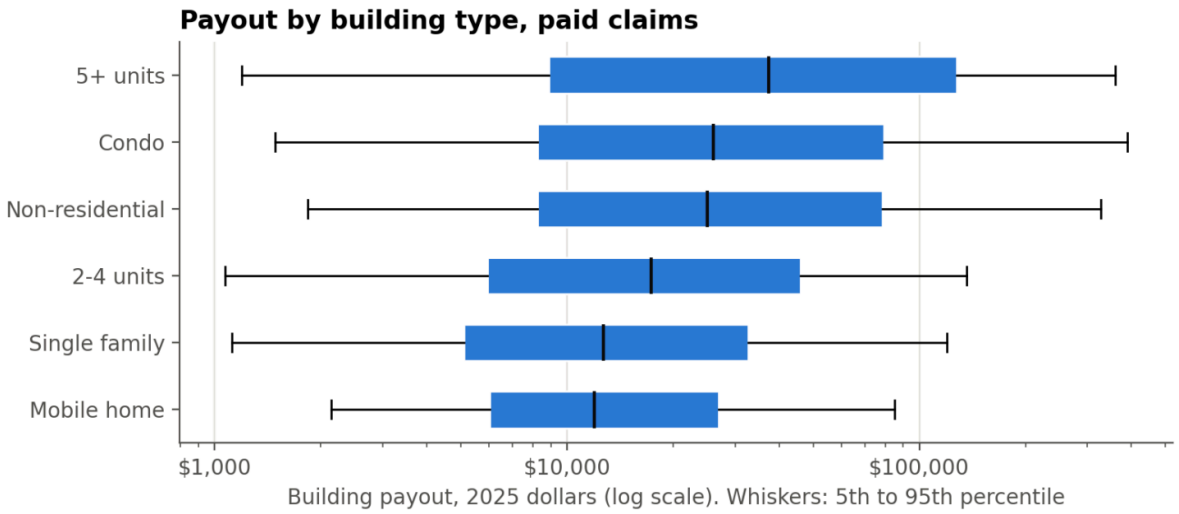
Payouts by flood zone



zone_group	claims	paid_rate_pct	median_payout	total_paid_millions
Low risk	20237	70.4	14923	513.8
Unknown	1671	62.7	13708	20.6
Zone A	77417	75.9	14033	2166.8
Zone V	10264	66.3	10491	203.1

Low-risk claims have the highest median payout, slightly above Zone A, and Zone V has the lowest. High-risk zones account for most claims and most total dollars, but that reflects the number of claims, not larger payouts per claim.

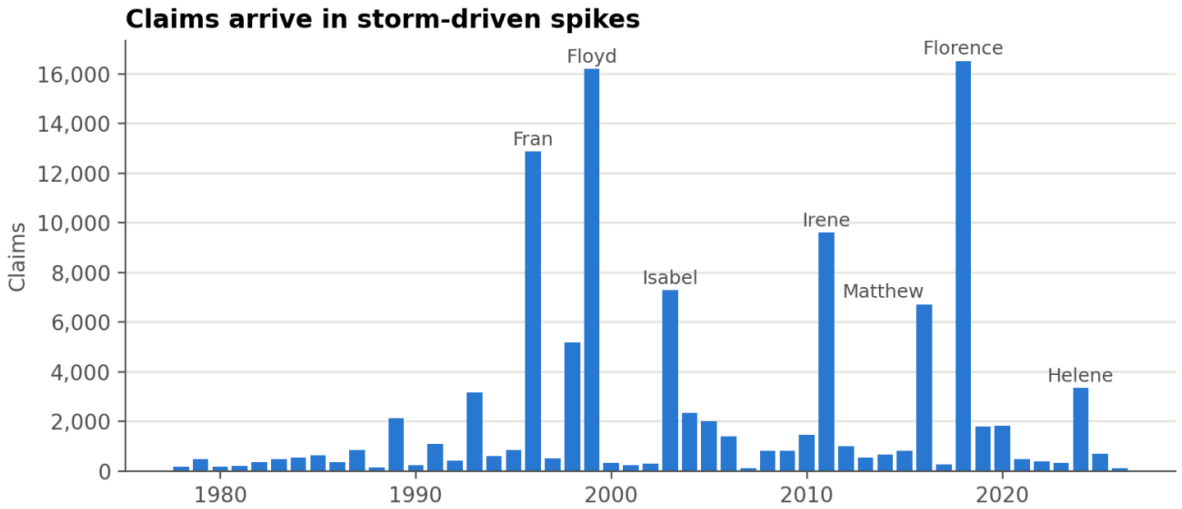
Payouts by building type



occupancy	claims	paid_rate_pct	median_payout	total_paid_millions
2-4 units	5387	68.9	17372	137.9
5+ units	3283	67.8	37290	213.9
Condo	902	72.5	26030	50.9
Mobile home	504	85.7	11932	10.1
Non-residential	9906	71.9	24987	516.4
Single family	89604	74.4	12678	1974.8
Unknown	3	100	5490	0.1

Larger buildings get larger payouts: 5+ unit buildings have about three times the median of single-family homes, which still make up most of the dollars paid.

Claims over time

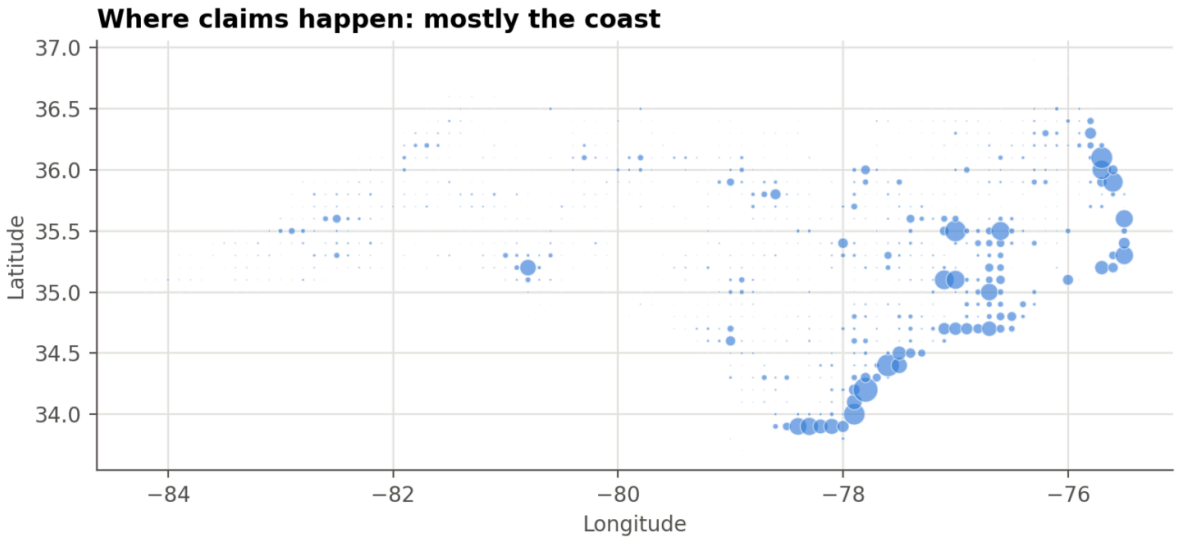


A handful of storms produce most claims, so claims from one storm are not independent. This matters for the tests and for how the models are validated.

event	claims	paid_rate_pct	median_payout	total_paid_millions
Hurricane Florence	15336	84.7	23173	693.1
Unnamed event	24050	67.2	9813	433.9
Hurricane Floyd	12334	73.5	15236	330.4
Hurricane Fran	10356	77.5	18069	312.3
Hurricane Matthew	5929	81.6	20247	233.3
Hurricane Irene	9337	78.7	13164	198.8
Hurricane Helene	2028	83.1	65415	190.2
Hurricane Isabel	6456	75.2	11684	142.8
Hurricane Dennis	3449	68.4	10833	49.6
Hurricane Bonnie	4378	58.1	8082	38.2

Florence is the costliest event. Helene has far fewer claims but by far the highest median payout, and it hit inland western NC rather than the coast.

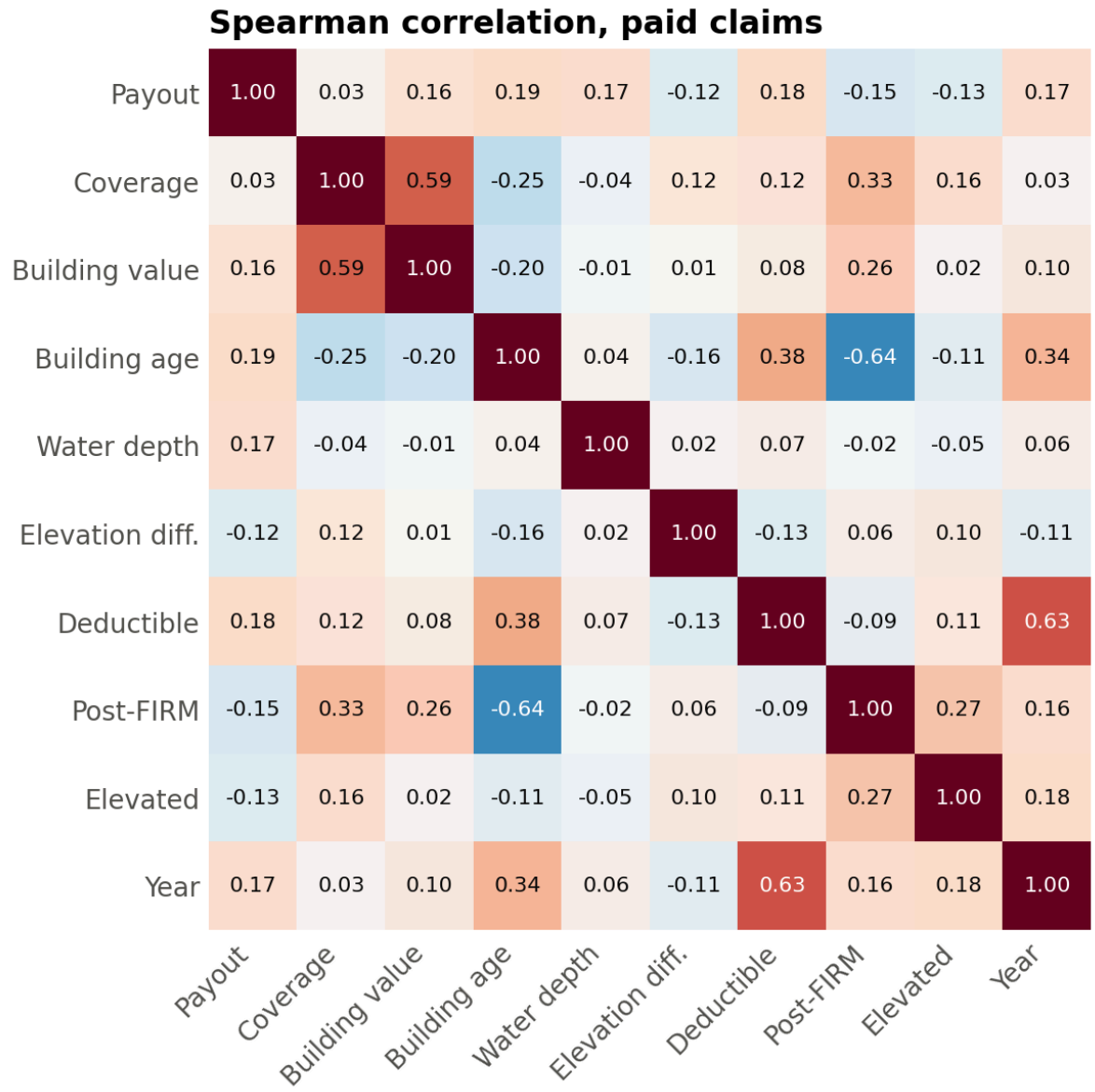
Where claims happen



county	claims	paid_rate_pct	median_payout	total_paid_millions
Dare	18949	70	9222	323
New Hanover	11183	71.6	16081	286.8
Craven	6126	85.4	20904	277
Beaufort	9462	83.7	11691	179.3
Carteret	8997	68	12095	159.4
Pender	5889	69.8	13141	155.4
Brunswick	9877	60.3	8693	145.3
Pamlico	4330	89.5	15259	115.8
Onslow	4431	67.9	12098	98.1
Buncombe	1192	69.5	41754	91.4

Claims concentrate along the coast and sounds, led by Dare and New Hanover. Buncombe is the inland exception: $65.5M of its $91.4M came from Helene.

Correlations



No single numeric column explains payouts: the strongest rank correlations with payout are below 0.2. Coverage is almost unrelated to payout, because coverage sets a ceiling and most claims are far below it. Deductible and year move together (0.63), so deductibles rose over time. Building age and post-FIRM are strongly linked (-0.64), so older buildings' results should not be read as two separate effects. Water depth is limited by its mixed units.

Do payouts change across policy periods?

period	claims	paid_rate_pct	median_payout	total_paid_millions	median_pa
1978-1994: before the 1994 reform act	11947	65.6	6804	152.7	
1995-2012: higher coverage limits	63197	71.4	12753	1368.1	
2013-2021: after Biggert-Waters	29593	81.2	18824	1107.3	
2022-2026: Risk Rating 2.0	4852	80	26755	276.1	

Median payout rises in every period, which at first looks like a policy effect. The last column tells a different story. Outside named storms, the median roughly doubled after 1994, when coverage limits were raised, then stayed flat at about $11,000 to $13,000. The later increases mostly reflect which storms hit, such as Florence and Helene, not policy changes. This is a comparison of medians, not proof of cause.

4. Statistical analysis

Four tests check the patterns seen in the EDA, each at the 0.05 level. Payouts are tested on the log scale because raw payouts are heavily skewed. Two cautions apply to all four. First, with tens of thousands of claims even tiny differences give very small p-values, so each test also reports an effect size, which measures how big the difference is. Second, claims from the same storm are not independent, which makes p-values look stronger than they are. The last part repeats the main test inside single storms to check this.

Test 1: Payout by flood zone (one-way ANOVA)

Null hypothesis: mean log payout is the same in low-risk zones, Zone A and Zone V.
Why this test: it compares the means of one numeric outcome across three groups. Log payouts are close to normal, and the groups have similar spreads (standard deviations 0.62 to 0.64).

F statistic	p-value	Eta squared
111.2	< 0.001	0.003

Tukey's test compares each pair of zones:

comparison	mean_log_diff	payout_ratio	p_value
Low risk vs Zone A	0.009	1.02	0.287
Low risk vs Zone V	0.127	1.34	< 0.001
Zone A vs Zone V	0.118	1.31	< 0.001

The null hypothesis is rejected, but zone explains only 0.3% of the variation in payouts (eta squared 0.003). Tukey's test shows where the difference is: low-risk and Zone A payouts are about the same (p = 0.29), and Zone V payouts are about 25% lower than both (the other two zones pay about 1.3 times as much). **Answer to the research question:** claims in high-risk zones do not get paid more. Zone A pays about the same as low-risk zones, and coastal Zone V pays less.

Test 2: Whether a claim is paid, by flood zone (chi-square)

Null hypothesis: the share of claims that get paid is the same in every zone group.
Why this test: both variables are categories (paid or not, and zone group), and every expected cell count is far above 5 (smallest 2,675).

Chi-square	p-value	Cramér's V
584	< 0.001	0.074

zone_group	paid rate (%)
Low risk	70.4
Zone A	75.9
Zone V	66.3

The null hypothesis is rejected. Zone A claims are paid most often (75.9%), then low-risk zones (70.4%), then Zone V (66.3%). The link is weak (Cramér's V 0.07), so zone says little about whether a claim gets paid.

Test 3: Post-FIRM vs pre-FIRM buildings (Welch t-test)

A FIRM is the flood map for a community. Post-FIRM buildings were built after the map existed, so they had to follow flood-zone building rules.
Null hypothesis: mean log payout is the same for post-FIRM and pre-FIRM buildings.
Why this test: it compares the means of two groups. The Welch version does not assume equal variances.

t statistic	p-value	Cohen's d
-40.4	< 0.001	-0.29

The null hypothesis is rejected. Post-FIRM buildings have a median payout of $10,485, compared with $16,821 for pre-FIRM buildings. The effect is small (d = -0.29). This matches the idea that newer flood rules reduce damage. But section 3 showed that post-FIRM buildings are also much newer (correlation -0.64), so the test cannot separate the effect of the building rules from building age.

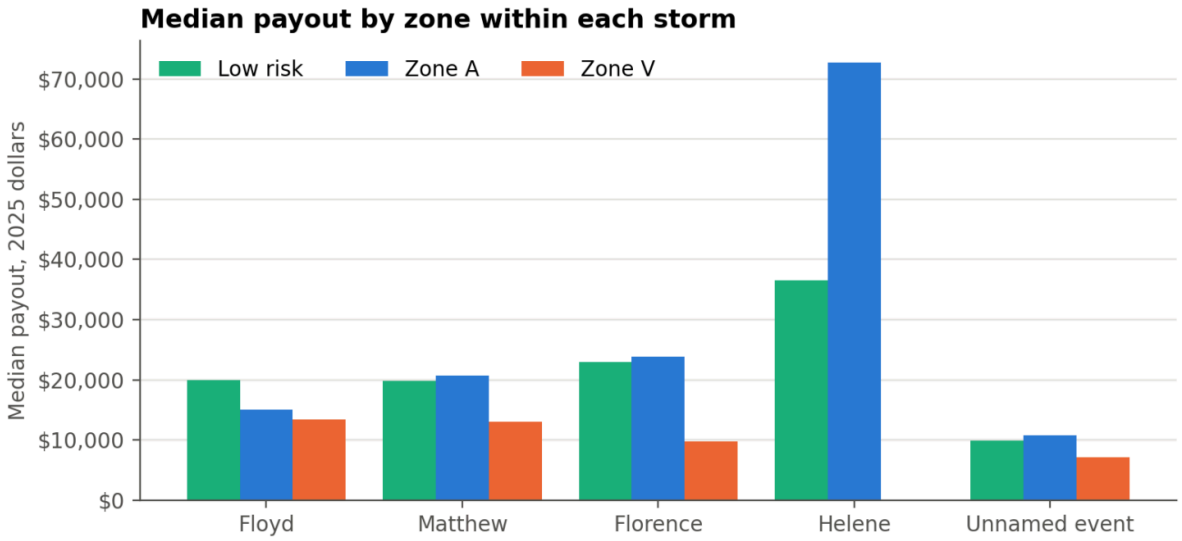
Test 4: Water depth and payout (Spearman correlation)

Null hypothesis: there is no monotonic relationship between water depth and payout.
Why this test: Spearman uses ranks, so it does not need normal data. That matters here because water depth mixes feet and inches, and ranks are less affected by the mix than raw values.

Spearman rho	p-value	Paid claims with depth
0.17	< 0.001	71,193

The null hypothesis is rejected: deeper water goes with larger payouts. The relationship is weak (rho 0.17), which is partly due to the mixed units. The true link is probably stronger than the data can show.

Check: zone comparison inside single storms



event	Low risk	Zone A	Zone V	eta_squared	p_value
Hurricane Floyd	19976	14996	13458	0.004	< 0.001
Hurricane Matthew	19826	20736	13038	0.005	< 0.001
Hurricane Florence	22993	23895	9738	0.014	< 0.001
Hurricane Helene	36529	72800	None	0.01	< 0.001
Unnamed event	9924	10730	7166	0.006	< 0.001

Repeating Test 1 inside each storm removes the storm mix and the dependence between storms. Zone V has the lowest median in every storm. Low-risk and Zone A are close in Matthew, Florence and unnamed events, but they differ in both directions elsewhere: low-risk claims paid about a third more in Floyd, and Zone A paid about twice as much in Helene (Helene had too few Zone V claims to show). Effect sizes stay small (eta squared 0.004 to 0.014). The main result holds: high-risk zones do not get larger payouts, and zone explains little of the variation. That points to building and flood details, which the models in section 5 use.

5. Predicting payouts

Two models predict the log of payout for paid claims: Linear Regression as the baseline and Random Forest as the more advanced model. The proposal also named XGBoost, but it was dropped to keep the project to standard scikit-learn models. Random Forest already covers a tree-based model.

Split by time, not at random: the models train on claims before Hurricane Florence (60,174 claims) and are tested on Florence and everything after it (20,669 claims, including Helene). A random split would put claims from the same storm in both sets, which makes the score look better than it really is. The time split asks the useful question: how well does the model do on storms it has never seen?

Features: coverage, building value, water depth, building age, deductible, location (latitude and longitude), post-FIRM, elevated, primary residence, rental, flood zone group, building type, number of floors and flood cause. Missing numbers are filled with the median, and categories are one-hot encoded.

Elevation difference was left out because half of it is missing. Year was left out because a Random Forest cannot predict beyond the years it was trained on.

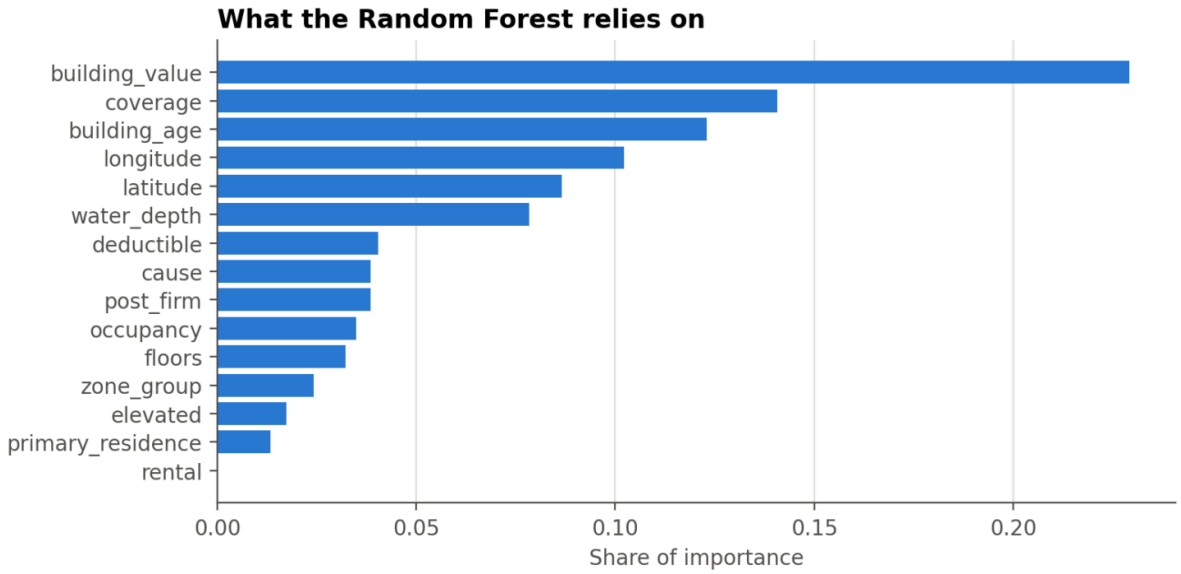
Results on the test storms

model	MAE	RMSE	R2	MAE_dollars	train_seconds
Median only (no model)	0.566	0.698	-0.116	45508.219	0
Linear Regression	0.528	0.653	0.022	46099.114	0.223
Random Forest	0.499	0.619	0.121	42027.742	3.509

MAE, RMSE and R^2 are on the log10 scale. A log MAE of 0.5 means a typical prediction is off by a factor of about 3. MAE in dollars is the average miss in 2025 dollars. The first row is a simple benchmark that predicts the training median for every claim.

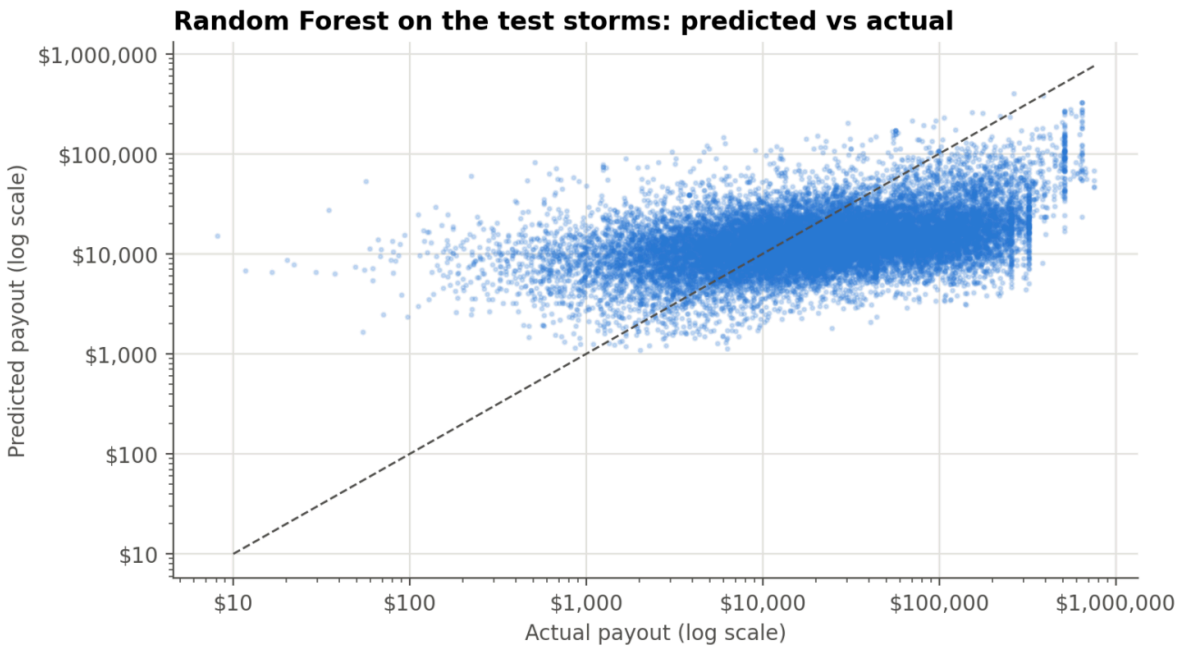
Random Forest does best, with R^2 of 0.12 against 0.02 for Linear Regression, but neither model predicts payouts well. Part of the reason is the time split: the test storms paid more (median $20,913 against $12,114 in training), which is why even the median benchmark scores below zero. The rest is missing information. The data has no measure of how much of the building was damaged, and water depth, the closest thing to one, mixes feet and inches. Random Forest trained in a few seconds, so compute time, listed as a challenge in the proposal, was not a problem at this size.

What the Random Forest relies on



Building value and coverage matter most, because they set how much there is to lose and the most that can be paid. Building age and location come next. Flood zone group ranks near the bottom, which agrees with section 4: the zone says little about how much a claim pays. Importance shows what the model uses, not what causes larger payouts.

Predicted vs actual



Points on the dashed line are perfect predictions. The model pulls everything toward the middle: it overpredicts small claims and underpredicts large ones. It can tell roughly whether a claim will be small or large, but it should not be used to estimate a single claim's payout.

6. AI-assisted analysis

The app includes an assistant built with LangChain and Google's Gemini API (model `gemini-3.5-flash`). A LangChain prompt template combines a fixed instruction, a summary of this project's results and the question, then sends it to Gemini and returns the answer as text. Every AI interaction in this project went through this workflow, including the decision log below.

Ask the assistant

Question about the analysis

What are the main limits of these results?

Ask

Decision log

Each major data preparation and analysis decision was put to the assistant through the workflow above (src/ai_log.py). For each one the log shows the prompt, the answer, an evaluation of the answer, and

whether the advice was accepted, modified or rejected.

Decision	AI advice
Missing values	Modified
Water depth units	Modified
Outliers	Modified
Inflation adjustment	Modified
Choice of statistical tests	Modified
Train and test split	Accepted
Model choice and features	Modified
Reading the model results	Accepted

<details>
<summary>Missing values: Modified</summary>

Prompt (gemini-3.5-flash, 2026-11-08)

Project results:
109,593 NC claims. buildingPropertyValue is missing on many claims; when it is missing only 2% of claims were paid, against 90% when it is present. floodWaterDuration is 94% missing for 2020s claims. foundationType holds 0 on 76% of rows, and the data dictionary does not define 0. Older claims lack deductible and flood zone.

Question: How should I handle these missing values? Should I drop incomplete rows, impute, or drop columns? The payout model will use paid claims only.

AI response

First, **filter your dataset to paid claims only**. This step alone will eliminate most claims with missing `buildingPropertyValue`, since 98% of claims missing that value were unpaid.

For the remaining paid claims, use a mixed strategy based on the results:

- `buildingPropertyValue`: Keep the column. Its presence strongly aligns with paid claims. After filtering for paid claims, keep the remaining rows.
- `foundationType`: Do not drop rows. Since 76% are `0`, treat `0` as a distinct "Unknown" category so you do not lose three-quarters of your dataset.
- `floodWaterDuration`: Drop this column if analyzing recent data, as 94% missingness in the 2020s makes imputation unreliable.
- **Deductible and Flood Zone**: Drop older rows or restrict your model's time window to newer years where these fields are populated.
</details>

Avoid dropping all incomplete rows outright, as you will lose critical data. (Note: The provided results do not test specific imputation algorithms, so you should experiment with using "missing" flags for numeric variables.)

Evaluation: Mostly sound. Keeping building value and dropping flood duration match the data. Two suggestions do not fit. Treating foundation type 0 as an Unknown category keeps a column that is 76% undefined, so it would carry almost no information. Dropping older rows that lack deductible or zone would remove whole eras, including Hurricane Floyd.

Decision (modified): Accepted keeping building value, dropping flood duration, and not dropping incomplete rows. Rejected the Unknown category for foundation type and dropped the column instead. Rejected dropping older rows: missing zones became an Unknown group and missing model inputs are filled with the median.

Water depth units: Modified

Prompt (gemini-3.5-flash, 2026-11-08)

Project results:
waterDepth has a median of 1 and the data dictionary says some values are in feet and some in inches, with no column saying which. Values of 900 and above are placeholders. It is 33% missing in the 2020s.

Question: Can I still use water depth to study payouts, and if so how? Should I try to convert the values to one unit?

AI response

Based on your project results:

Can you convert the values to one unit? No. The results explicitly state there is no column indicating which records are in feet and which are in inches. Without knowing the original unit for each row, you cannot reliably convert them to a single unit.

Can you still use it to study payouts, and how? The current results cannot support a specific method for using `waterDepth` to study payouts due to these data quality issues:

- **Unclear units:** A value of "1" could mean 1 inch or 1 foot (the median is 1).
- **Placeholders:** Values of 900 and above are not actual measurements and must be removed or filtered out.
- **Missing data:** 33% of the records in the 2020s are missing entirely.

If you choose to use this variable, you must first filter out the placeholder values (≥ 900) and account for the missing 2020s data. However, because you cannot distinguish feet from inches, the results do not show a way to reliably analyze its direct relationship with insurance payouts.

Evaluation: Right that the values cannot be converted without a unit column and that 900+ values must go. Too cautious in saying depth cannot be used at all. A rank-based method uses only the order of values, so it is less affected by the unit mix than methods that use the raw numbers.

Decision (modified): Removed 900+ values and did not convert units, as suggested. Kept depth, using Spearman rank correlation in section 4 and as a model input, and stated the mixed units as a limit that weakens the measured relationship.

Outliers: Modified

Prompt (gemini-3.5-flash, 2026-11-08)

Project results:
80,843 paid claims in 2025 dollars: median $13,822, mean $35,924, max $2,735,296. On a log scale the 1.5 IQR rule flags 50 claims above $703,514 and 637 below $291. Separately, 336 payments are larger than the listed building coverage.

Question: Should I remove these outliers, and what should I do with the payments that exceed coverage?

AI response

Based on the project results provided, there is **not enough information to state whether you should remove the outliers or how to handle payments exceeding coverage**.

Here is what the results confirm:

- **Log-Scale Outliers:** The 1.5 IQR rule flags 50 high claims (above $703,514) and 637 low claims (below $291). However, the results do not show whether these are data entry errors or valid, extreme flood losses.
- **Payments Exceeding Coverage:** The data confirms 336 cases where payments exceed building coverage limits. However, the results only identify that these cases exist; they do not explain why this happened (e.g., policy adjustments or data mismatch).

Because the results only flag these statistical anomalies and do not provide data entry context or policy rules, **the results alone cannot support a decision to delete or alter these records.**

To decide what to do, you cannot rely solely on these summary results; you would need to investigate the specific records and NFIP reporting guidelines further.

Evaluation: It gave no recommendation, only a statement that the summary cannot decide. The useful part is the caution that the flags do not show errors. It missed the domain point that large flood losses are real and are what insurance covers.

Decision (modified): Kept all outliers, as nothing shows they are errors, and used a log scale to limit their effect. The 336 payments above coverage stay in and are reported as a data quality limit, since the data does not explain them.

Inflation adjustment: Modified

Prompt (gemini-3.5-flash, 2026-11-08)

Project results:
Claims run from 1978 to 2025. I plan to convert payouts to 2025 dollars using the annual average CPI-U (FRED series CPIAUCNS).

Question: Is CPI-U a good choice for adjusting flood payouts, or is there a better index? What should I say about it in the report?

AI response

CPI-U is a reasonable baseline for general inflation, but it tracks standard consumer goods rather than structural repairs. For building claims, a construction-specific index—such as the Producer Price Index (PPI) for residential construction inputs or a construction cost index—is often better because material and labor costs frequently rise at different rates than general consumer prices.

What to say in your report:

- **Methodology:** Explain that you used CPI-U (FRED: CPIAUCNS) to adjust payouts from 1978 to 2025 dollars based on general economic purchasing power.
- **Limitation:** Note that CPI-U may not perfectly reflect trends in construction and repair costs, which could lead to under- or over-estimating historical building damages.

Result note: The provided project details only state your timeframe and intended index; they do not contain data comparing different indices. Therefore, the current project results cannot support a claim about which index produces the most accurate trend for your specific North Carolina dataset.

Evaluation: A fair point: CPI-U tracks consumer prices, not repair costs, and a construction index could move differently. It also correctly says the project has no data to show which index is better.

Decision (modified): Kept CPI-U because it covers the full 1978 to 2025 range with one simple annual series and the proposal named it. Added its suggested limitation to the report: repair costs may have risen at a different rate.

Choice of statistical tests: Modified

Prompt (gemini-3.5-flash, 2026-11-08)

Project results:
Research question: do claims in high-risk flood zones get paid more? Zones grouped as Low risk

(20,237 claims), Zone A (77,417) and Zone V (10,264). Log10 payout is close to normal with similar spreads (SD 0.62 to 0.64). A few storms (Floyd, Matthew, Florence, Helene) produce most of the claims.

Question: Which test should I use to compare payouts across the three zones, and what should I watch out for with this many claims coming from a few storms?

AI response

To compare the average log10 payouts across the three flood risk zones, you should use a **one-way ANOVA (Analysis of Variance)**. Your data meets the main requirements for ANOVA: you are comparing three groups, and the log10 payouts are roughly normal with nearly identical spreads (SDs between 0.62 and 0.64).

What to watch out for: The main issue is **lack of independence (clustering)**. Because a few major storms (Floyd, Matthew, Florence, and Helene) produced most of the claims, the data points are not completely independent. Claims from the same storm share common factors like storm intensity, local geography, and regional damage.

Standard ANOVA assumes every claim is an independent event. Ignoring this storm clustering can artificially inflate your sample size, shrink your standard errors, and lead to false-positive results (finding a significant difference where one might not exist).

To fix this, you should watch out for spatial/temporal dependencies and consider methods that account for storm-level effects, such as a mixed-effects model or using clustered standard errors by storm.

Evaluation: ANOVA is the right test and the clustering warning is the key issue. Mixed-effects models and clustered standard errors go beyond this course. It also missed the second half of the question: with tens of thousands of claims, p-values will be tiny even for trivial differences.

Decision (modified): Used one-way ANOVA with Tukey's test. Instead of a mixed model, repeated the ANOVA inside each storm, which removes the storm mix. Added effect sizes (eta squared, Cramér's V, Cohen's d) to every test to handle the large-sample problem the answer left out.

Train and test split: Accepted

Prompt (gemini-3.5-flash, 2026-11-08)

Project results:
Model target is log10 payout on paid claims. Claims cluster in storms. Option 1: a random 80/20 split. Option 2: train on claims before Hurricane Florence (2018-09-01, 60,174 claims) and test on Florence and later (20,669 claims, including Helene).

Question: Which split should I use to evaluate the payout models, and why?

AI response

You should use **Option 2** (the temporal split).

Here is why:

- **Claims cluster in storms:** Flood claims naturally occur in groups tied to major weather events.
- **Prevents data leakage:** A random 80/20 split (Option 1) would put claims from the *same storm* into both training and test sets. This creates data leakage, making the model look better than it actually is.
- **Simulates real-world testing:** Option 2 trains the model on past claims and tests it on future, unseen storms (like Florence and Helene). This accurately measures how well your model predicts payouts for new storm events, which is how it would be used in practice.

Evaluation: Correct and well reasoned. A random split would put claims from the same storm on both sides and make the scores look better than they are.

Decision (accepted): Used the time split: train before Florence, test Florence onward.

Model choice and features: Modified

Prompt (gemini-3.5-flash, 2026-11-08)

Project results:
The proposal named Linear Regression, Random Forest and XGBoost. The course needs one baseline and one advanced model, and I want to keep to standard scikit-learn. Candidate features: coverage, building value, water depth, building age, deductible, latitude, longitude, post-FIRM, elevated, primary residence, rental, zone group, occupancy, floors, flood cause, year of loss, and elevation difference (51% missing).

Question: Which models should I keep, and are there features I should leave out given the time-based split?

AI response

Based on your course requirements and preference for standard scikit-learn:

- **Models to keep:** Keep **Linear Regression** as your baseline model and **Random Forest** as your advanced model. Both are standard scikit-learn models, whereas XGBoost requires a separate external library.
- **Feature selection:** The provided results note that **elevation difference** has high missingness (51% missing), which makes it a strong candidate to leave out. However, the provided results **do not contain information about a time-based split** or how it impacts feature selection. Because of this, I cannot recommend leaving out specific features (such as `year of loss`) based on a time-based split using only these results.

Evaluation: The model advice is right. It said it could not judge features against a time split, which misses a real problem: every test year is later than every training year, and a Random Forest cannot predict outside the range it was trained on.

Decision (modified): Kept Linear Regression and Random Forest and dropped XGBoost. Left out elevation difference, as suggested. Also left out year of loss, for the reason the answer missed.

Reading the model results: Accepted

Prompt (gemini-3.5-flash, 2026-11-08)

Project results:
Test set (Florence onward). Median benchmark R2 -0.12. Linear Regression MAE 0.528, RMSE 0.653, R2 0.02 (log10 scale). Random Forest MAE 0.499, RMSE 0.619, R2 0.12. Test median payout $20,913 vs $12,114 in training. Top Random Forest importances: building value 0.23, coverage 0.14, building age 0.12, then longitude, latitude, water depth. Zone group is near the bottom.

Question: How should I interpret these results? Can I conclude that flood zone does not affect payouts?

AI response

Here is how to interpret your results:

1. Model Performance and Shift The models struggle to predict claim sizes accurately. The best model (Random Forest) explains only 12% of the variance ($R^2 = 0.12$). This difficulty is driven by a major shift in the data: post-Florence test claims were much more severe, with a median payout ($20,913) nearly double the training median ($12,114).

2. Key Drivers The primary predictors of payout size are financial and structural attributes (building value, policy coverage, building age), followed by precise location (latitude/longitude) and water depth.

Can you conclude flood zone does not affect payouts? No, you cannot.

"Zone group" ranking near the bottom of feature importance only means it provided little extra predictive value to *this specific model*. Other variables—like exact coordinates and water depth—likely captured the geographic flood risk better than broad zone categories. Because the overall model fit is weak ($R^2 = 0.12$), low feature importance in the model does not prove a lack of effect in the real world.

Evaluation: A good reading. Low importance in a weak model is not proof of no effect, and the shift between training and test storms explains part of the low R^2. One word overstates: "more severe" test claims. The data shows they were paid more, not that the damage was worse.

Decision (accepted): The conclusions rest the zone finding on the section 4 tests, including the within-storm check, not on feature importance. They describe the test storms as paying more, not as

more severe.

Across the eight decisions the advice was accepted twice and modified six times. The assistant was most useful on standard method choices, such as the test and the train/test split. It was weakest where the answer depended on knowing the data or the domain: it was too cautious about water depth and outliers, and it missed both the large-sample p-value problem and the reason year cannot be a feature.

7. Conclusions

Do high-risk zones get paid more? No. Among paid claims, Zone A pays about the same as low-risk zones, and coastal Zone V pays about 25% less. Zone V also has the lowest median inside every storm, so the result is not just a mix of storms. Zone explains well under 2% of the variation in payouts.
What predicts payouts? Building value and coverage, which set how much can be lost and paid, then building age, location and water depth. Post-FIRM buildings are paid less, but that cannot be separated from building age.
How well can payouts be predicted? Not well. The Random Forest reaches R^2 0.12 on storms it has not seen, against 0.02 for Linear Regression. It can tell small claims from large ones roughly, but not estimate a single claim.
Over time: median payouts rose, but outside named storms they have been flat since the 1994 reform. The rise mostly reflects which storms hit.

Limits. Water depth mixes feet and inches. Building value is missing mostly on unpaid claims. CPI-U may not track repair costs. The data has no measure of how much of a building was damaged, which is likely the biggest missing driver. Claims within a storm are not independent.
Next steps. Add storm-level data such as rainfall or surge height, and test whether a damage measure, where available, raises the model's accuracy.